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(54) LEG EXTENSION MACHINE

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(57) ABSTRACT

A leg extension machine includes a frame portion supported by the bottom surface and standing upright; a seating portion that is connected to the frame portion and provides a space for a user to sit and the standing angle and the height of which are adjustable; an exercise rotation portion connected to the frame portion, disposed adjacent to the seating portion to provide a space for a seated user to hang his or her legs, and rotated by an external force applied by the user; a weight portion supported by the frame portion and accommodating a plurality of weights; and a weight delivering portion connected to the frame portion, connecting the exercise rotation portion and the weight portion, and linked to the rotation of the exercise rotation portion to raise and lower the weight of the weight portion.

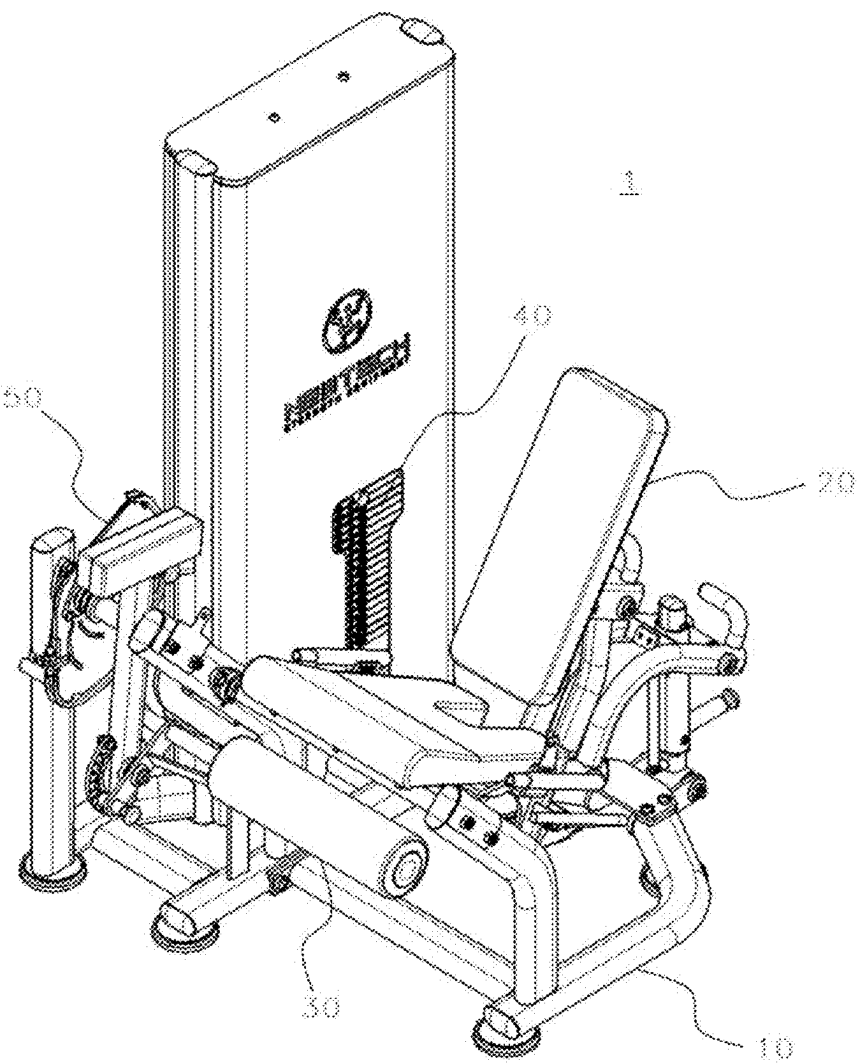


FIG. 1

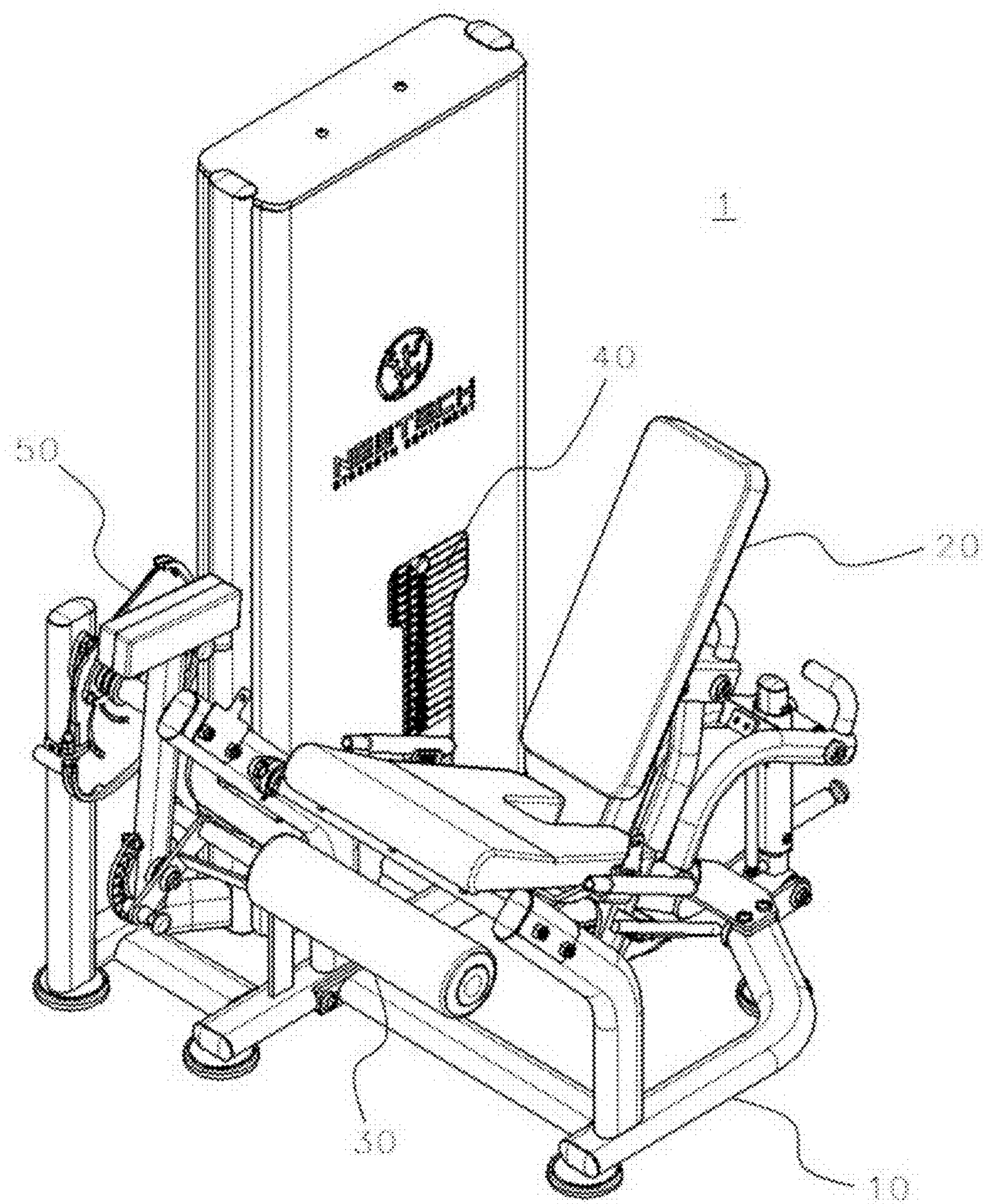


FIG. 2

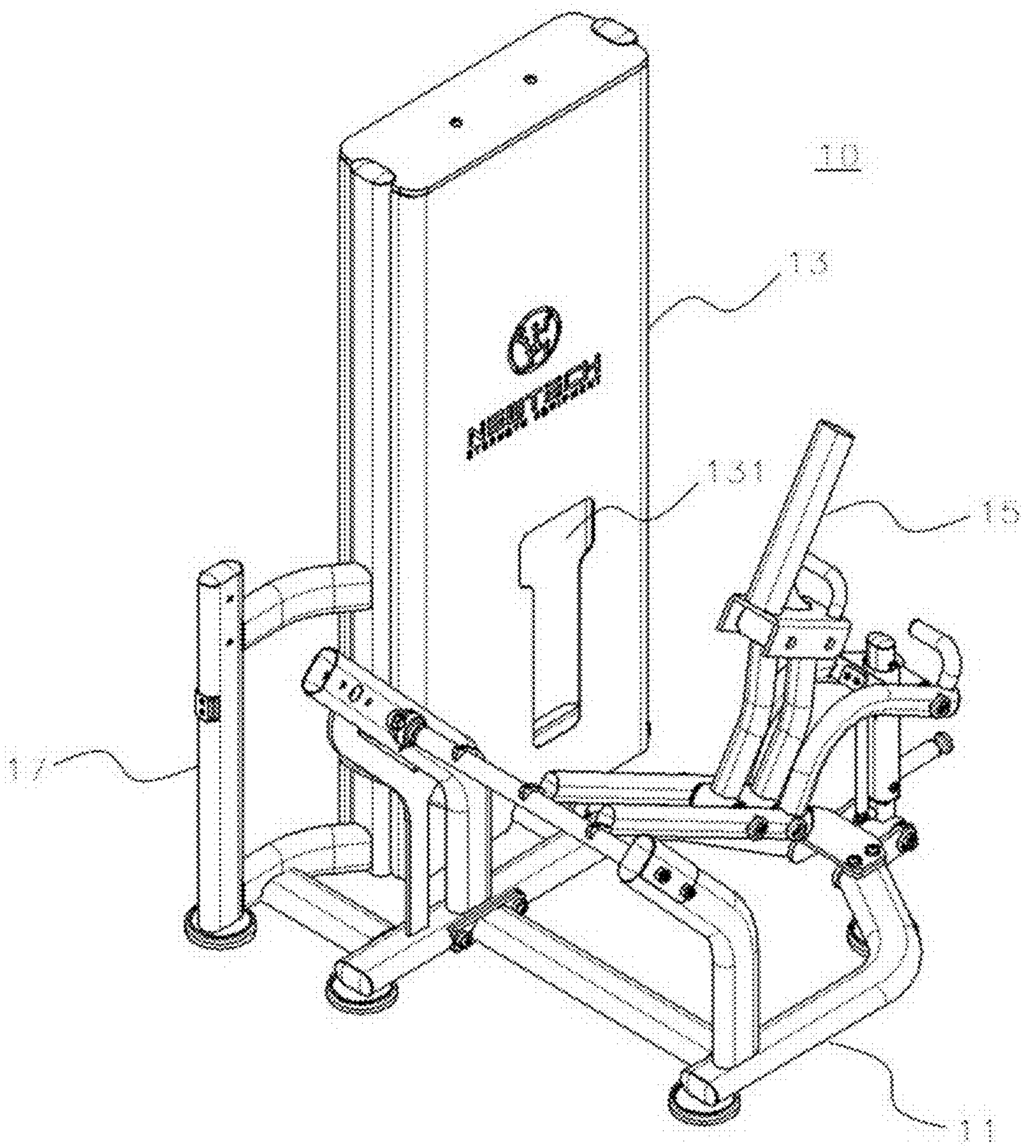


FIG. 3

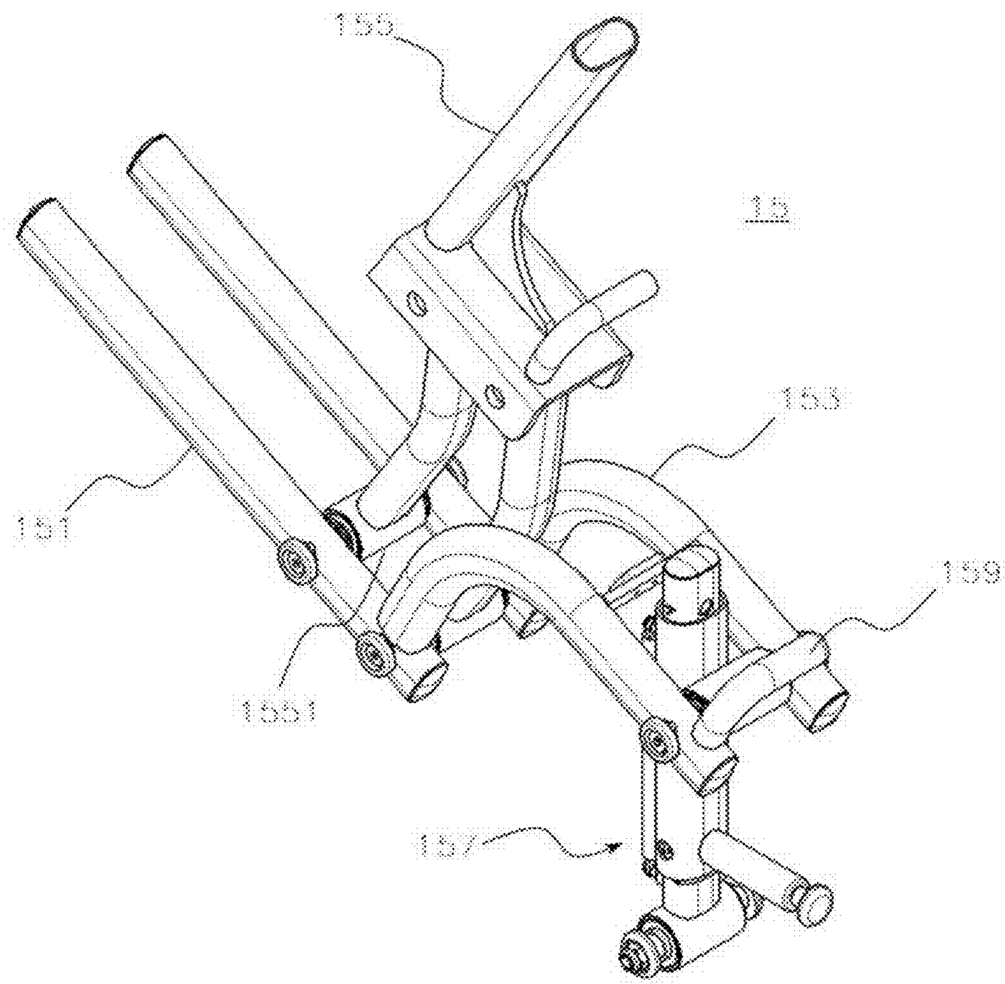


FIG. 4

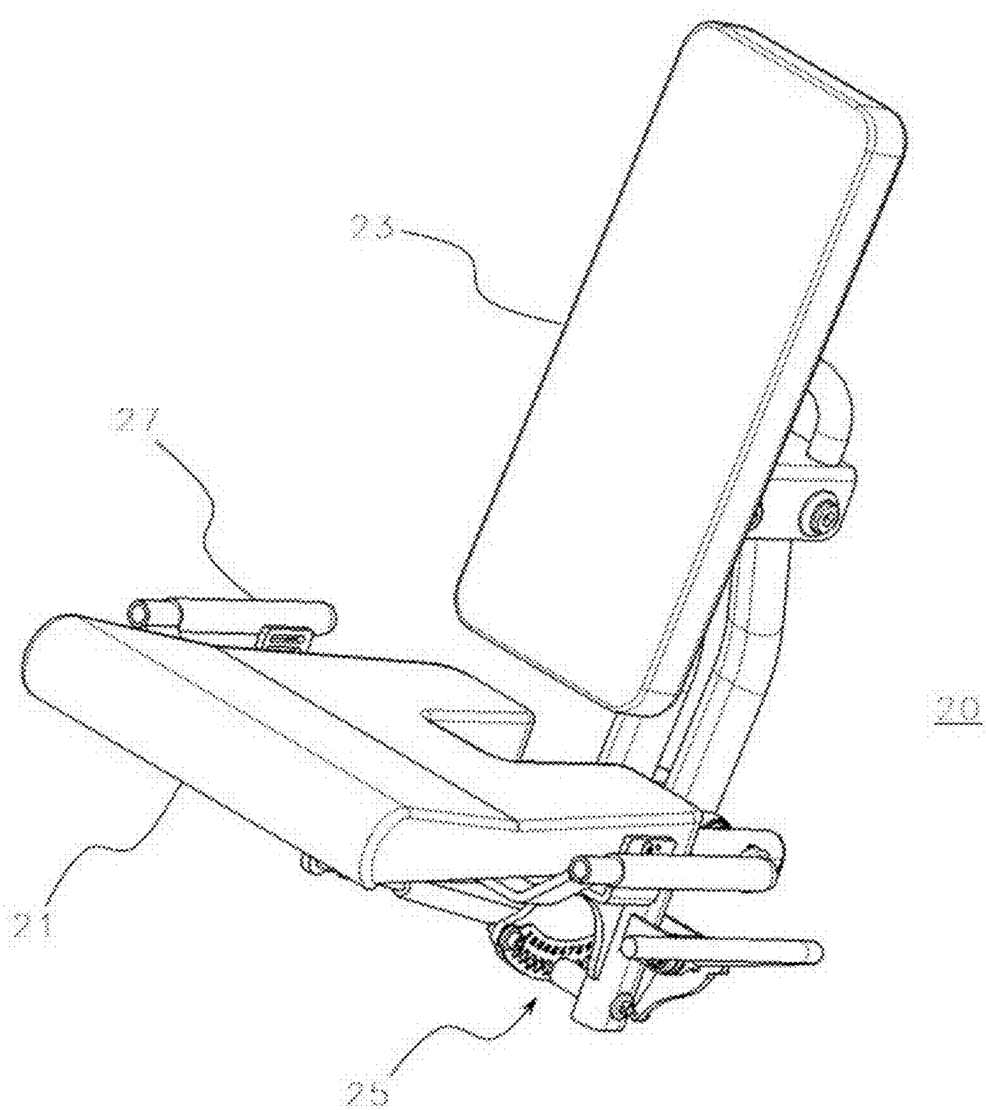


FIG. 5

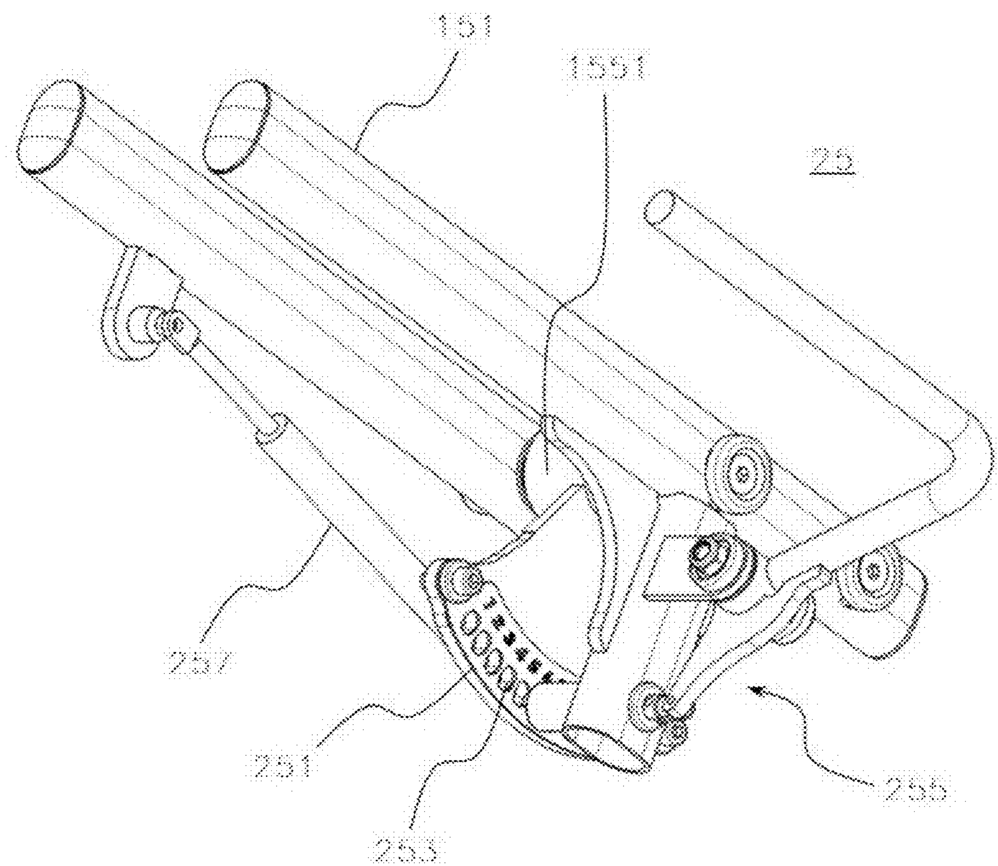


FIG. 6

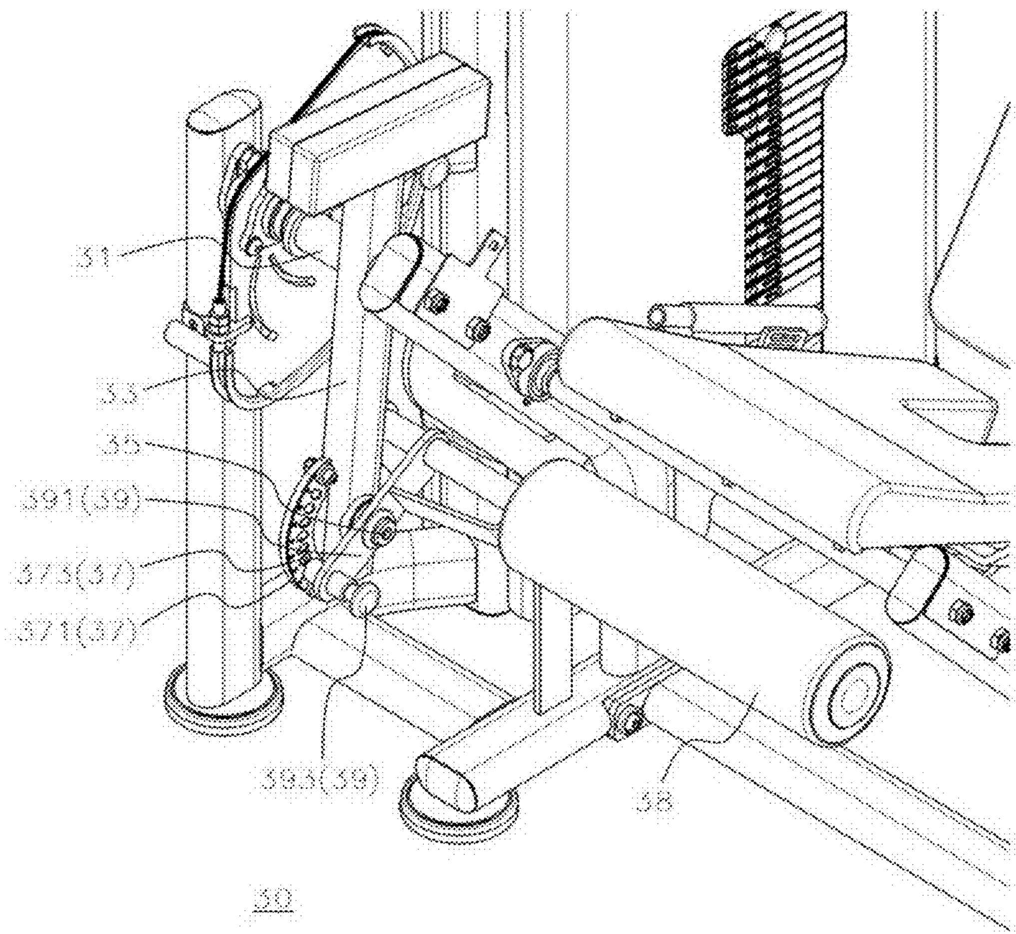


FIG. 7

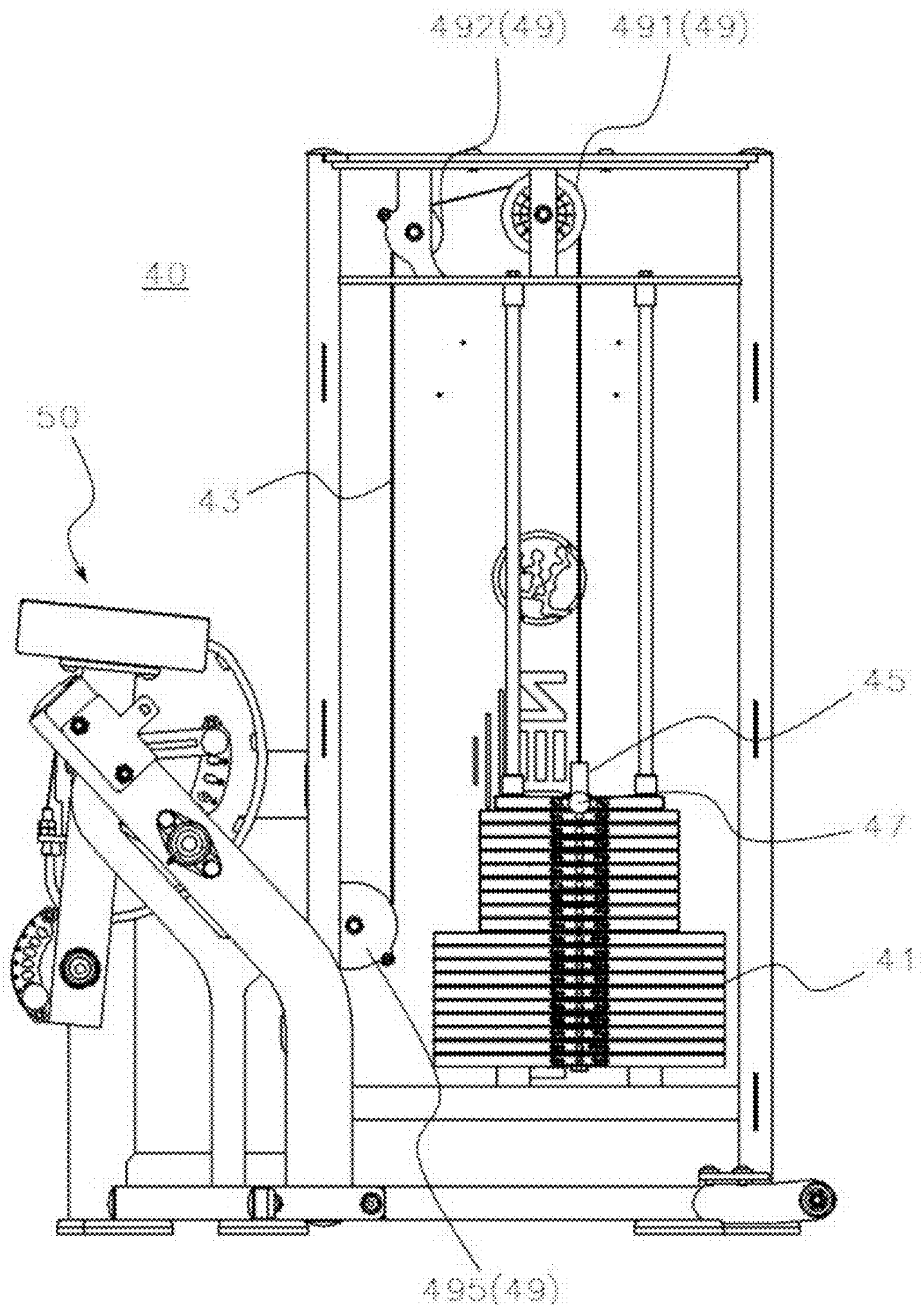


FIG. 8

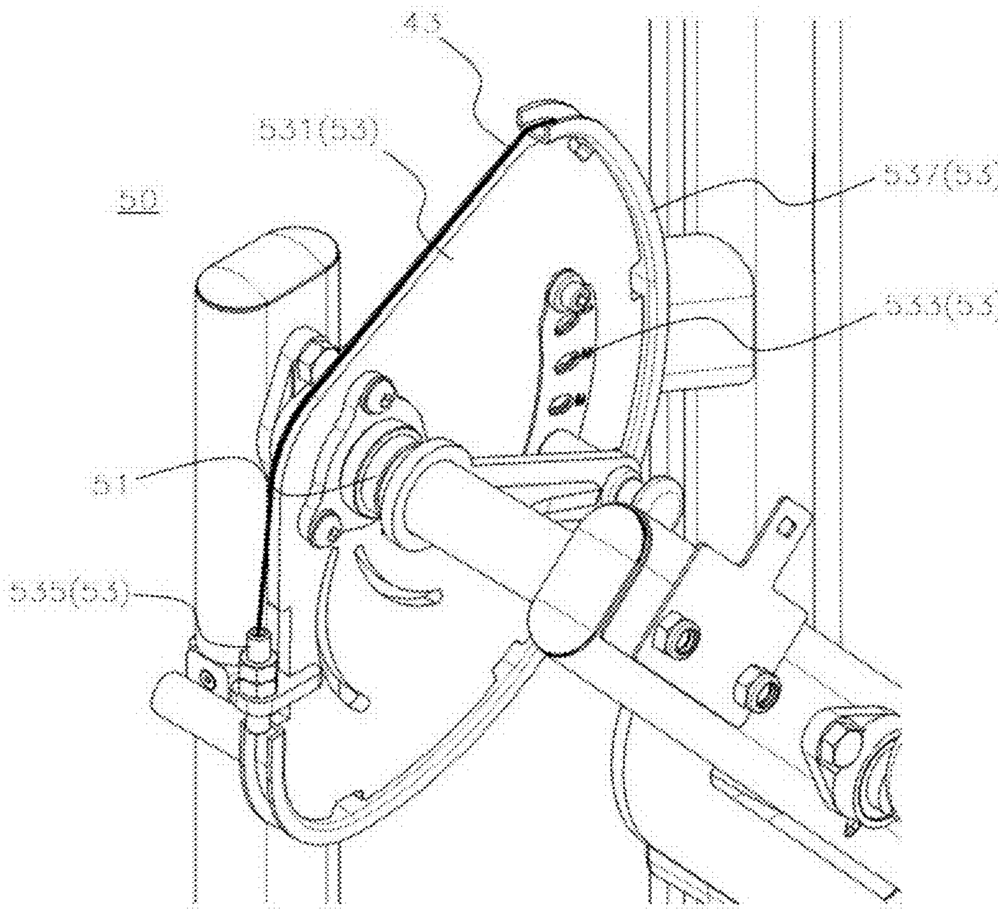
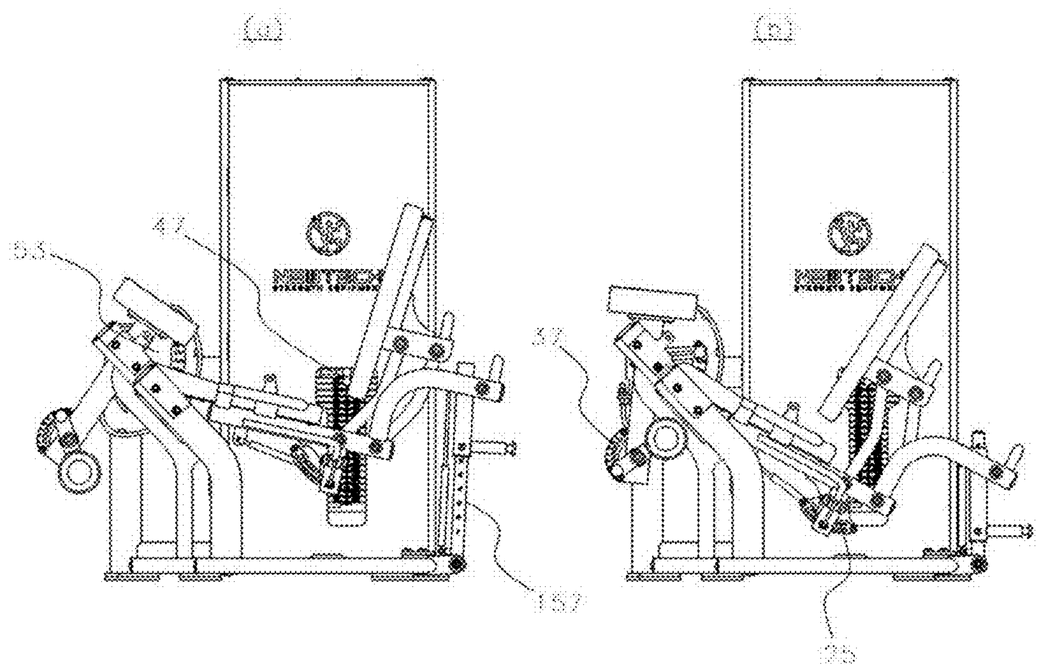


FIG. 9



LEG EXTENSION MACHINE

BACKGROUND OF THE INVENTION

[0001] The present disclosure relates to a leg extension machine, and, more particularly, to a leg extension machine allowing a seated user to work out by raising a lever connected to a weight to lift the weight and adjust the machine to suit not only the length but also the thickness of the user's legs.

[0002] Recently, as interest in prevention of obesity or adult diseases and self-management has increased, a lot of people of all ages and genders have enjoyed a variety of types of exercise. There may be various types of exercise for that purpose, but, in recent days, many people sign up for gyms to exercise.

[0003] A number of people use gyms because they are equipped with a variety of exercise equipment and allow users to appropriately adjust weight loads for muscular work outs to suit their individual physical condition and amount of exercise. In addition, at gyms, it is possible to exercise without restrictions due to season, weather, and time.

[0004] In general, exercises available at gyms can be largely divided into aerobic exercise and machine exercise, and can be further divided into upper body muscle exercise, lower body muscle exercise, abdominal muscle exercise, etc., depending on the purpose of exercise.

[0005] Among them, examples of representative exercise equipment for lower body muscle exercise include leg curl machines, leg extension machines, aerial walking machines, etc.

[0006] In the case of a conventional leg extension machine, it is possible to adjust the position of a seat to suit the length of a user's legs, but it is impossible to adjust the machine to fit the thickness of the user's thighs, causing inconvenience in use.

[0007] Therefore, research on a leg extension machine that can be adjusted to not only the length but also the thickness of a user's legs is required.

RELATED ART LITERATURE

Patent Literature

[0008] (Patent Literature 1) Korea Utility Model No. 20-0453493

SUMMARY OF THE INVENTION

[0009] The present disclosure has been made in an effort to provide a leg extension machine allowing a seated user to work out by raising a lever connected to a weight to lift the weight and adjust the machine to suit not only the length but also the thickness of the user's legs.

[0010] In addition, the present disclosure has been made in an effort to provide a leg extension machine designed to be compact not only to reduce production costs but also to provide convenience in maintenance and management.

[0011] A leg extension machine according to an embodiment of the present disclosure may include a frame portion supported by the bottom surface and standing upright; a seating portion that is connected to the frame portion and provides a space for a user to sit and the standing angle and the height of which are adjustable; an exercise rotation portion connected to the frame portion, disposed adjacent to the seating portion to provide a space for a seated user to hang his or her legs, and rotated by an external force applied by the user; a weight portion supported by the frame portion and accommodating a plurality of weights; and a weight delivering portion connected to the frame portion, connecting the exercise rotation portion and the weight portion, and linked to the rotation of the exercise rotation portion to raise and lower the weight of the weight portion. The seating portion may include a hip seating portion supporting the buttocks of a seated user; a back seating portion supporting the back of a seated user; a seating angle adjuster connected to the frame portion and the back seating portion to be used to adjust the angle of the back seating portion; and a plurality of grip portions connected to both sides of the hip seating portion and arranged in parallel with the hip seating portion for a user to grip them.

[0012] The frame portion according to an embodiment of the present disclosure may include a lower support portion supported by the bottom surface; a weight accommodating portion disposed at the rear of the seating portion, standing up from the lower support portion, and having an internal space to accommodate the weight portion; a seating support portion standing up from the lower support portion and supporting the seating portion; a delivering support portion standing up from the lower support portion and disposed adjacent to the weight accommodating portion to support the weight delivering portion; and an exercise support portion connecting the delivering support portion and the seating support portion and supporting the exercise rotation portion. The seating support portion may include a pair of seating support pieces supporting the hip seating portion; a pair of bending support pieces extending from the first point of the pair of seating support pieces; a rotating support piece connecting the pair of seating support pieces and the back seating portion, where a plurality of rotational links are provided in the space between the pair of seating support pieces so that the back seating portion rotates; and a seating height adjuster arranged to adjust the height of the pair of bending support pieces and having one end link-connected to one point of the lower support portion and the other end link-connected to the pair of bending support pieces.

[0013] The seating angle adjuster according to an embodiment of the present disclosure may include an angle adjusting plate fixed to the rotational link, operating in conjunction with the back seating portion, and formed in a fan shape; a plurality of angle adjusting holes penetrating the angle adjusting plate and spaced apart from each other along an arc shape; a fixing lever module coupled to the second point of the seating support piece and inserted into at least one of the plurality of angle adjusting holes to fix the angle of the back seating portion; and a damper connecting the third point of the seating support piece and the first point of the angle adjusting plate and pulling the angle adjusting plate toward the third point of the seating support piece when the fixing lever module is released.

[0014] The weight portion according to an embodiment of the present disclosure may include a weight stacking portion accommodated in the weight accommodating portion and formed by stacking a plurality of weights separable from each other; a wire having one side connected to the weight delivering portion and the other side connected to the weight stacking portion and raising and lowering the weight stacking portion based on the movement of the weight delivering portion; a weight mediating portion connecting the wire and the weight stacking portion and allowing a selected subordinate weight to vary according to how the weight stacking portion is connected; an intermediate pin determining how the weight mediating portion and the weight stacking portion are connected to each other; and a guide roller providing a movement path for the wire while being accommodated in the weight accommodating portion. The guide roller may include a first upper roller piece disposed on the upper side within the weight accommodating portion and disposed on the same line as the weight mediating portion; a second upper roller piece accommodated in the weight accommodating portion, disposed adjacent to the first upper roller piece, and disposed biased toward the weight delivering portion; and a lower roller piece accommodated in the weight accommodating portion, disposed below the second upper roller piece, and disposed adjacent to the weight delivering portion. The weight delivering portion may include a delivering rotation axis connected rotatably to the delivering support portion; and a delivering control plate connected to and interlocked with the delivering rotation axis and rotated by the wire. The delivering control plate may include a first fan-shaped portion formed in a fan shape; a plurality of first through holes penetrating the first fan-shaped portion and spaced apart from each other along an arc shape; a wire fixing portion formed on one side of the first fan-shaped portion to fix the wire; and a wire guide portion guiding the wire fixed to the wire fixing portion to be disposed along the outer peripheral surface of the first fan-shaped portion.

[0015] The exercise rotation portion according to an embodiment of the present disclosure may include a connection rotation axis formed in concentric multi-stages to surround the outer peripheral surface of the delivering rotation axis and connected thereto to rotate around the delivering rotation axis; an interlocking bar connected perpendicularly to the connection rotation axis; an adjustment rotation axis connected rotatably to the end of the interlocking bar; an exercise control plate formed at the end of the interlocking bar; and an exercise connection portion connected to the adjustment rotation axis, providing a space where a resting portion, on which a user's legs can rest, is mounted, and connected detachably to the exercise control plate. The connection rotation axis may include a bending selection portion disposed parallel to the first fan-shaped portion and having both ends disposed on top of the plurality of first through holes; and an interlocking button portion formed at the end of the bending selection portion, inserted into one of the plurality of first through holes and then released from the through hole by an applied external force, and returned to be inserted into an opposing through hole when the applied external force is released. Depending on the position where the interlocking button portion is connected to the first fan-shaped portion, the angle between the bottom surface and the interlocking bar may change, changing the resting portion's height from the bottom surface. The exercise control plate may include a second fan-shaped portion formed in a fan shape; and a plurality of second through holes penetrating the second fan-shaped portion and spaced apart from each other along an arc shape. The exercise connection portion may include an exercise selection portion disposed parallel to the second fan-shaped portion and having both ends disposed on top of the plurality of second through holes; and an exercise button portion formed at the end of the exercise selection portion, inserted into any one of the plurality of second through holes and then released from the through hole by an applied external force, and returned to be inserted into an opposing through hole when the applied external force is released. Depending on the position at which the exercise button portion is connected to the second fan-shaped portion, the distance between the resting portion and the seating portion may be changed.

[0016] The leg extension machine according to one embodiment of the present disclosure may allow a seated user to work out by raising a lever connected to a weight to lift the weight and adjust the machine to suit not only the length but also the thickness of the user's legs.

[0017] In addition, the leg extension machine according to one embodiment of the present disclosure may be designed to be compact not only to reduce production costs but also to provide convenience in maintenance and management.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. **1** is a view for illustrating a leg extension machine according to an embodiment of the present disclosure.

[0019] FIG. **2** is a view for illustrating a frame portion according to an embodiment of the present disclosure.

[0020] FIG. **3** is a view for illustrating a seating support portion according to an embodiment of the present disclosure.

[0021] FIG. **4** is a view for illustrating a seating portion according to an embodiment of the present disclosure.

[0022] FIG. **5** is a view for illustrating a seating angle adjuster according to an embodiment of the present disclosure.

[0023] FIG. **6** is a view for illustrating an exercise rotation portion according to an embodiment of the present disclosure.

[0024] FIG. **7** is a view for illustrating a weight portion according to an embodiment of the present disclosure.

[0025] FIG. **8** is a view for illustrating a weight delivering portion according to an embodiment of the present disclosure.

[0026] FIG. **9** is a view for comparing the leg extension machines with different angles and heights according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0027] Hereinafter, the specific embodiments of the present disclosure will be described in detail with reference to the drawings. However, the technology of the present disclosure is not limited to the presented embodiments. A person having ordinary in the art who understands the technology of the present disclosure would be able to easily suggest other embodiments within the technology of other regressive disclosures or the present disclosure by adding, changing, or removing components within the technology of the present disclosure, but such embodiments can also be said to be within the technology of the present disclosure.

[0028] Hereinafter, a leg extension machine **1** according to the present disclosure will be described in detail with reference to the attached FIGS. **1** to **8**.

[0029] FIG. **1** is a view for illustrating the leg extension machine according to an embodiment of the present disclosure.

[0030] Referring to FIG. **1**, the leg extension machine **1** according to an embodiment of the present disclosure may include a frame portion **10**, a seating portion **20**, an exercise rotation portion **30**, a weight portion **40**, and a weight delivering portion **50**.

[0031] The frame portion **10** may be supported by the floor and stand upright. The frame portion **10** will be examined in more detail with reference to FIG. **2**.

[0032] FIG. **2** is a view for illustrating the frame portion according to an embodiment of the present disclosure.

[0033] Referring to FIG. **2**, the frame portion **10** according to an embodiment of the present disclosure may include a lower support portion **11**, a weight accommodating portion **13**, a seating support portion **15**, and a delivering support portion **17**.

[0034] The lower support portion **11** may be supported by the floor.

[0035] The weight accommodating portion **13** may be disposed at the rear of the seating portion **20**, stand up from the lower support portion **11**, and have an internal space to accommodate the weight portion **40**. The weight accommodating portion **13** may include an adjustment opening **131** that is open on the front side facing the seating portion **20** to expose at least a portion of the weight portion **40** to the outside.

[0036] The seating support portion **15** may stand up from the lower support portion **11** and support the seating portion **20**. The seating support portion **15** will be examined in more detail with reference to FIG. **3**.

[0037] FIG. **3** is a view for illustrating the seating support portion according to an embodiment of the present disclosure.

[0038] Referring to FIG. **3**, the seating support portion **15** according to an embodiment of the present disclosure may include a pair of seating support pieces **151**, a pair of bending support pieces **153**, a rotating support piece **155**, and a seating height adjuster **157**.

[0039] The pair of seating support pieces **151** may support a hip seating portion **21**.

[0040] The pair of bending support pieces **153** may extend from the first point of the pair of seating support pieces **151**.

[0041] The rotating support piece **155** may connect the pair of seating support pieces **151** and a back seating portion **23**, and a plurality of rotational links **1551** may be provided in the space between the pair of seating support pieces **151** so that the back seating portion **23** can rotate.

[0042] One end of the seating height adjuster **157** for adjusting the height of the pair of bending support pieces **153** may be link-connected to one point of the lower support portion **11**, and the other end thereof may be link-connected to the pair of bending support pieces **153**. The seating height adjuster **157** may include two bars and a height adjustment button. In order to easily adjust the height with the seating height adjuster **157**, a height adjustment grip portion **159** may be provided on one point of the bending support piece **153**. Because the seating height adjuster **157** may be provided, it may be possible to adjust the angle of the hip seating portion **21** by adjusting the height of one side of the hip seating portion **21**.

[0043] Referring back to FIG. **2**, the delivering support portion **17** may stand up from the lower support portion **11** and may be disposed adjacent to the weight accommodating portion **13** to support the weight delivering portion **50**.

[0044] Referring back to FIG. **1**, the seating portion **20** may be connected to the frame portion **10** and provide a space for a user to sit, and it may be possible to adjust the angle at which the seating portion **20** stands up and the height thereof. The seating portion **20** will be examined in more detail with reference to FIG. **4**.

[0045] FIG. **4** is a view for illustrating the seating portion according to an embodiment of the present disclosure.

[0046] Referring to FIG. **4**, the seating portion **20** according to an embodiment of the present disclosure may include the hip seating portion **21**, the back seating portion **23**, a seating angle adjuster **25**, and a pair of grip portions **27**.

[0047] The hip seating portion **21** may support the buttocks of a seated user.

[0048] The back seating portion **23** may support the back of a seated user.

[0049] The seating angle adjuster **25** may be connected to the frame portion **10** and the back seating portion **23** and used to adjust the angle of the back seating portion **23**. The seating angle adjuster **25** will be examined in more detail with reference to FIG. **5**.

[0050] FIG. **5** is a view for illustrating the seating angle adjuster according to an embodiment of the present disclosure.

[0051] Referring to FIG. **5**, the seating angle adjuster **25** according to one embodiment of the present disclosure may include an angle adjusting plate **251**, a plurality of angle adjusting holes **253**, a fixing lever module **255**, and a damper **257**.

[0052] The angle adjusting plate **251** fixed to the rotational link **1551** may operate in conjunction with the back seating portion **23** and may be formed in a fan shape.

[0053] The plurality of angle adjusting holes **253** may penetrate the angle adjusting plate **251** and may be spaced apart from each other along an arc shape.

[0054] The fixing lever module **255** may be coupled to the second point of the seating support piece **151**, and may be inserted into at least one of the plurality of angle adjusting holes **253** to fix the angle of the back seating portion **23**.

[0055] The damper **257** may connect the third point of the seating support piece **151** and the first point of the angle adjusting plate **251**, and may pull the angle adjusting plate **251** toward the third point of the seating support piece **151** when the fixing lever module **255** is released.

[0056] Referring back to FIG. **4**, the pair of grip portions **27** may be connected to both sides of the hip seating portion **21**, and may be arranged parallel to the hip seating portion **21** so that a user can grip them.

[0057] Referring back to FIG. **1**, the exercise rotation portion **30** may be connected to the frame portion **10** and disposed adjacent to the seating portion **20** to provide a space for a seated user to hang his or her legs, and may be rotated by an external force applied by the user. The exercise rotation portion **30** will be examined in more detail with reference to FIG. **6**.

[0058] FIG. **6** is a view for illustrating the exercise rotation portion according to an embodiment of the present disclosure.

[0059] Referring to FIG. **6**, the exercise rotation portion **30** according to an embodiment of the present disclosure may include a connection rotation axis **31**, an interlocking bar **33**, an adjustment rotation axis **35**, an exercise control plate **37**, and an exercise connection portion **39**.

[0060] The connection rotation axis **31** may be formed in concentric multi-stages to surround the outer peripheral surface of a delivering rotation axis **51** and may be connected thereto to rotate around the delivering rotation axis

51. The connection rotation axis **31** may include a bending selection portion **311** and an interlocking button portion **313**.

[0061] The bending selection portion **311** may be disposed parallel to a first fan-shaped portion **531**, and its ends may be disposed on top of a plurality of first through holes **533**.

[0062] The interlocking button portion **313** may be formed at the end of the bending selection portion **311**. The interlocking button portion **313** may be inserted into one of the plurality of first through holes **533** and then released from the through hole by an applied external force, and may be returned to be inserted into an opposing through hole when the applied external force is released. Depending on the position where the interlocking button portion **313** is connected to the first fan-shaped portion **531**, the angle between the bottom surface and the interlocking bar **33** may change, changing a resting portion **38**'s height from the bottom surface.

[0063] Specifically, the interlocking button portion **313** may allow the knee height to be adjusted based on a user's body type or body condition. Usually, it may be difficult for users who cannot extend their knees properly to hang their legs on the resting portion **38** while lying down, and, in this case, the users can work out by lifting the resting portion **38**.

[0064] The interlocking bar **33** may be connected perpendicularly to the connection rotation axis **31**.

[0065] The adjustment rotation axis **35** may be rotatably connected to the end of the interlocking bar **33**.

[0066] The exercise control plate **37** may be formed at the end of the interlocking bar **33**. The exercise control plate **37** may include a second fan-shaped portion **371** formed in a fan shape, and a plurality of second through holes **373** penetrating the second fan-shaped portion **371** and spaced apart from each other along an arc shape.

[0067] The exercise connection portion **39** may be connected to the adjustment rotation axis **35**, may provide a space where the resting portion **38**, on which a user's legs can rest, is mounted, and may be detachably connected to the exercise control plate **37**. The exercise connection portion **39** may include an exercise selection portion **391** disposed parallel to the second fan-shaped portion **371** and having both ends disposed on top of the plurality of second through holes **373** and an exercise button portion **393** formed at the end of the exercise selection portion **391**. The exercise button portion **393** may be inserted into any one of the plurality of second through holes **373** and then released from the through hole by an applied external force, and may be returned to be inserted into an opposing through hole when the applied external force is released. Depending on the position at which the exercise button portion **393** is connected to the second fan-shaped portion **371**, the distance between the resting portion **38** and the seating portion **20** may be changed.

[0068] Referring back to FIG. **1**, the weight portion **40** may be supported by the frame portion **10** and may accommodate a plurality of weights. The weight portion **40** will be examined in more detail with reference to FIG. **7**.

[0069] FIG. **7** is a view for illustrating the weight portion according to an embodiment of the present disclosure.

[0070] Referring to FIG. **7**, the weight portion **40** according to an embodiment of the present disclosure may include a weight stacking portion **41**, a wire **43**, a weight mediating portion **45**, an intermediate pin **47**, and a guide roller **49**.

[0071] The weight stacking portion **41** may be accommodated in the weight accommodating portion **13** and may be formed by stacking a plurality of weights separable from each other.

[0072] One side of the wire **43** may be connected to the weight delivering portion **50**, and the other side thereof may be connected to the weight stacking portion **41**. The wire **43** may raise and lower the weight stacking portion **41** based on the movement of the weight delivering portion **50**.

[0073] The weight mediating portion **45** may connect the wire **43** and the weight stacking portion **41**, and may allow a selected subordinate weight to vary according to how the weight stacking portion **41** is connected.

[0074] The intermediate pin **47** may determine how the weight mediating portion **45** and the weight stacking portion **41** are connected to each other.

[0075] The guide roller **49** may provide a movement path for the wire **43** while being accommodated in the weight accommodating portion **13**.

[0076] The guide roller **49** may include a first upper roller piece **491**, a second upper roller piece **492**, and a lower roller piece **495**.

[0077] The first upper roller piece **491** may be disposed on the upper side within the weight accommodating portion **13**, and may be disposed on the same line as the weight mediating portion **45**.

[0078] The second upper roller piece **492** may be accommodated in the weight accommodating portion **13**, may be disposed adjacent to the first upper roller piece **491**, and may be disposed biased toward the weight delivering portion **50**.

[0079] The lower roller piece **495** may be accommodated in the weight accommodating portion **13**, disposed below the second upper roller piece **492**, and disposed adjacent to the weight delivering portion **50**.

[0080] The guide roller **49** may maintain the wire **43** bouncy without being twisted between the weight stacking portion **41** and the weight delivering portion **50**, thereby increasing the efficiency with which an external force is delivered.

[0081] Referring back to FIG. **1**, the weight delivering portion **50** may be connected to the frame portion **10**, may connect the exercise rotation portion **30** and the weight portion **40**, and may be linked to the rotation of the exercise rotation portion **30** to raise and lower the weight of the weight portion **40**. The weight delivering portion **50** will be examined in more detail with reference to FIG. **8**.

[0082] FIG. **8** is a view for illustrating the weight delivering portion according to an embodiment of the present disclosure.

[0083] Referring to FIG. **8**, the weight delivering portion **50** according to an embodiment of the present disclosure may include the delivering rotation axis **51** and a delivering control plate **53**.

[0084] The delivering rotation axis **51** may be rotatably connected to the delivering support portion **17**.

[0085] The delivering control plate **53** may be connected to and interlocked with the delivering rotation axis **51** and may be rotated by the wire **43**.

[0086] The delivering control plate **53** may include the first fan-shaped portion **531**, the plurality of first through holes **533**, a wire fixing portion **535**, and a wire guide portion **537**.

[0087] The first fan-shaped portion **531** may be formed in a fan shape.

[0088] The plurality of first through holes may penetrate the first fan-shaped portion **531** and may be spaced apart from each other along an arc shape.

[0089] The wire fixing portion **535** may be formed on one side of the first fan-shaped portion **531** to fix the wire **43**.

[0090] The wire guide portion **537** may guide the wire **43** fixed to the wire fixing portion **535** to be disposed along the outer peripheral surface of the first fan-shaped portion **531**.

[0091] The wire guide portion **537** may prevent the wire **43** from being separated from the first fan-shaped portion **531** when the wire **43** is wound or unwound along the outer peripheral surface of the first fan-shaped portion **531**.

[0092] In addition, hereinafter, an example in which the angle of the leg extension machine is adjusted will be examined with reference to FIG. **9**.

[0093] FIG. **9** is a view for comparing the leg extension machines with different angles and heights according to an embodiment of the present disclosure.

[0094] As shown in FIG. **9**, it may be possible for users to work out effectively by adjusting the seating height adjuster **157**, the seating angle adjuster **25**, the exercise control plate **37**, the intermediate pin **47**, the delivering control plate **53**, etc. to fit each user's body. The leg extension machine **1** according to the present disclosure can be used according to various embodiments in addition to the embodiments shown in figures (a) and (b) of FIG. **9**.

[0095] As examined above, according to one embodiment of the present disclosure, it may be possible for a seated user to work out by raising a lever connected to a weight to lift the weight and adjust the leg extension machine to suit not only the length but also the thickness of the user's legs.

[0096] In addition, because the leg extension machine according to one embodiment of the present disclosure may be designed to be compact, it may be possible to not only reduce manufacturing costs but also easily maintain and manage the exercise equipment.

[0097] One embodiment of the present disclosure has been described with limited examples and drawings, but is not limited to the above-described examples. In addition, a person having ordinary skill in the art to which the present disclosure pertains can make various modifications and variations from this description. Therefore, one embodiment of the present disclosure should be understood only by the claims set forth below, and all equivalent modifications thereof can be said to fall within the scope of the technology of the present disclosure.

1. A leg extension machine comprising:
- a frame portion supported by the bottom surface and standing upright;
- a seating portion that is connected to the frame portion and provides a space for a user to sit and the standing angle and the height of which are adjustable;
- an exercise rotation portion connected to the frame portion, disposed adjacent to the seating portion to provide a space for a seated user to hang his or her legs, and rotated by an external force applied by the user;
- a weight portion supported by the frame portion and accommodating a plurality of weights; and
- a weight delivering portion connected to the frame portion, connecting the exercise rotation portion and the weight portion, and linked to the rotation of the exercise rotation portion to raise and lower the weight of the weight portion,

wherein the seating portion includes:
- a hip seating portion supporting the buttocks of a seated user;
- a back seating portion supporting the back of a seated user;
- a seating angle adjuster connected to the frame portion and the back seating portion to be used to adjust the angle of the back seating portion; and
- a plurality of grip portions connected to both sides of the hip seating portion and arranged in parallel with the hip seating portion for a user to grip them.

2. The leg extension machine of claim **1**, wherein the frame portion includes:
- a lower support portion supported by the bottom surface;
- a weight accommodating portion disposed at the rear of the seating portion, standing up from the lower support portion, and having an internal space to accommodate the weight portion;
- a seating support portion standing up from the lower support portion and supporting the seating portion;
- a delivering support portion standing up from the lower support portion and disposed adjacent to the weight accommodating portion to support the weight delivering portion; and
- an exercise support portion connecting the delivering support portion and the seating support portion and supporting the exercise rotation portion, and

the seating support portion includes:
- a pair of seating support pieces supporting the hip seating portion;
- a pair of bending support pieces extending from the first point of the pair of seating support pieces;
- a rotating support piece connecting the pair of seating support pieces and the back seating portion, where a plurality of rotational links are provided in the space between the pair of seating support pieces so that the back seating portion rotates; and
- a seating height adjuster arranged to adjust the height of the pair of bending support pieces and having one end link-connected to one point of the lower support portion and the other end link-connected to the pair of bending support pieces.

3. The leg extension machine of claim **2**, wherein the seating angle adjuster includes:
- an angle adjusting plate fixed to the rotational link, operating in conjunction with the back seating portion, and formed in a fan shape;
- a plurality of angle adjusting holes penetrating the angle adjusting plate and spaced apart from each other along an arc shape;
- a fixing lever module coupled to the second point of the seating support piece and inserted into at least one of the plurality of angle adjusting holes to fix the angle of the back seating portion; and
- a damper connecting the third point of the seating support piece and the first point of the angle adjusting plate and pulling the angle adjusting plate toward the third point of the seating support piece when the fixing lever module is released.

4. The leg extension machine of claim **3**, wherein the weight portion includes:
- a weight stacking portion accommodated in the weight accommodating portion and formed by stacking a plurality of weights separable from each other;

a wire having one side connected to the weight delivering portion and the other side connected to the weight stacking portion and raising and lowering the weight stacking portion based on the movement of the weight delivering portion;
a weight mediating portion connecting the wire and the weight stacking portion and allowing a selected subordinate weight to vary according to how the weight stacking portion is connected;
an intermediate pin determining how the weight mediating portion and the weight stacking portion are connected to each other; and
a guide roller providing a movement path for the wire while being accommodated in the weight accommodating portion,
the guide roller includes:
a first upper roller piece disposed on the upper side within the weight accommodating portion and disposed on the same line as the weight mediating portion;
a second upper roller piece accommodated in the weight accommodating portion, disposed adjacent to the first upper roller piece, and disposed biased toward the weight delivering portion; and
a lower roller piece accommodated in the weight accommodating portion, disposed below the second upper roller piece, and disposed adjacent to the weight delivering portion,
the weight delivering portion includes:
a delivering rotation axis connected rotatably to the delivering support portion; and
a delivering control plate connected to and interlocked with the delivering rotation axis and rotated by the wire, and
the delivering control plate includes:
a first fan-shaped portion formed in a fan shape;
a plurality of first through holes penetrating the first fan-shaped portion and spaced apart from each other along an arc shape;
a wire fixing portion formed on one side of the first fan-shaped portion to fix the wire; and
a wire guide portion guiding the wire fixed to the wire fixing portion to be disposed along the outer peripheral surface of the first fan-shaped portion.

5. The leg extension machine of claim **4**, wherein the exercise rotation portion includes:
a connection rotation axis formed in concentric multi-stages to surround the outer peripheral surface of the delivering rotation axis and connected thereto to rotate around the delivering rotation axis;
an interlocking bar connected perpendicularly to the connection rotation axis;
an adjustment rotation axis connected rotatably to the end of the interlocking bar;
an exercise control plate formed at the end of the interlocking bar; and
an exercise connection portion connected to the adjustment rotation axis, providing a space where a resting portion, on which a user's legs can rest, is mounted, and connected detachably to the exercise control plate,
the connection rotation axis includes:
a bending selection portion disposed parallel to the first fan-shaped portion and having both ends disposed on top of the plurality of first through holes; and
an interlocking button portion formed at the end of the bending selection portion, inserted into one of the plurality of first through holes and then released from the through hole by an applied external force, and returned to be inserted into an opposing through hole when the applied external force is released,
depending on the position where the interlocking button portion is connected to the first fan-shaped portion, the angle between the bottom surface and the interlocking bar changes, changing the resting portion's height from the bottom surface,
the exercise control plate includes:
a second fan-shaped portion formed in a fan shape; and
a plurality of second through holes penetrating the second fan-shaped portion and spaced apart from each other along an arc shape,
the exercise connection portion includes:
an exercise selection portion disposed parallel to the second fan-shaped portion and having both ends disposed on top of the plurality of second through holes; and
an exercise button portion formed at the end of the exercise selection portion, inserted into any one of the plurality of second through holes and then released from the through hole by an applied external force, and returned to be inserted into an opposing through hole when the applied external force is released, and
depending on the position at which the exercise button portion is connected to the second fan-shaped portion, the distance between the resting portion and the seating portion is changed.

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